Emotions and visual awareness by attentional factors (attentional unawareness)

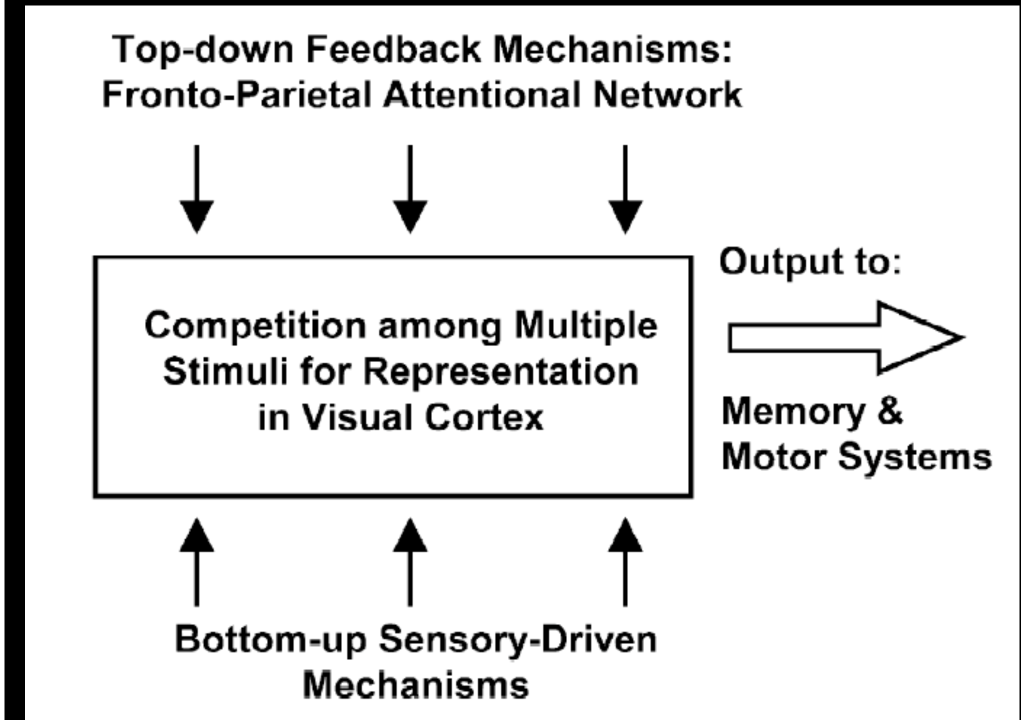
Attentional Unawareness

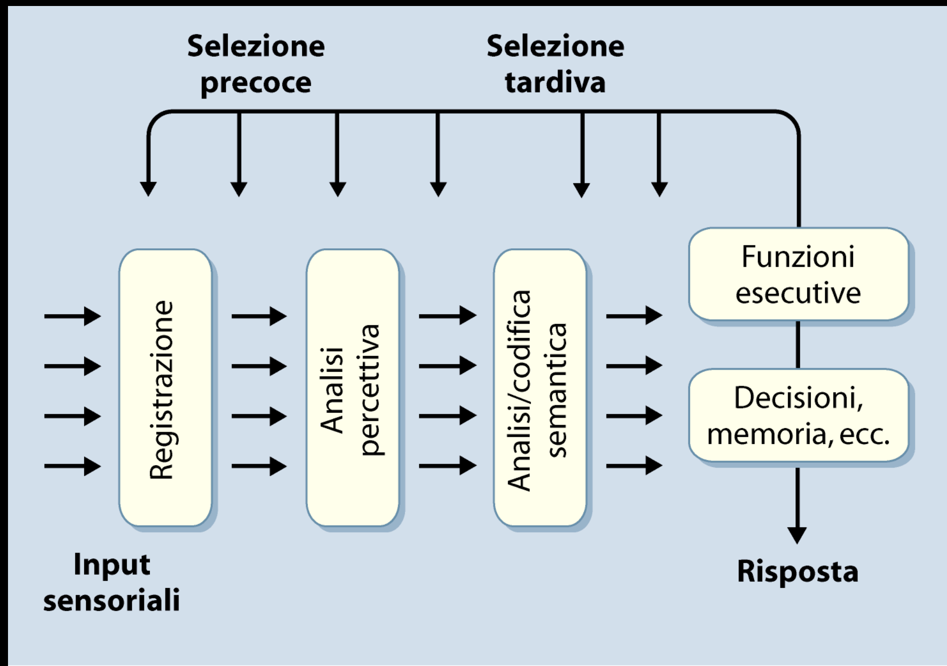
- Is potentially accessible to consciousness
- But fails to be consciously perceived because attention is allocated elsewhere

Sensory Unawareness

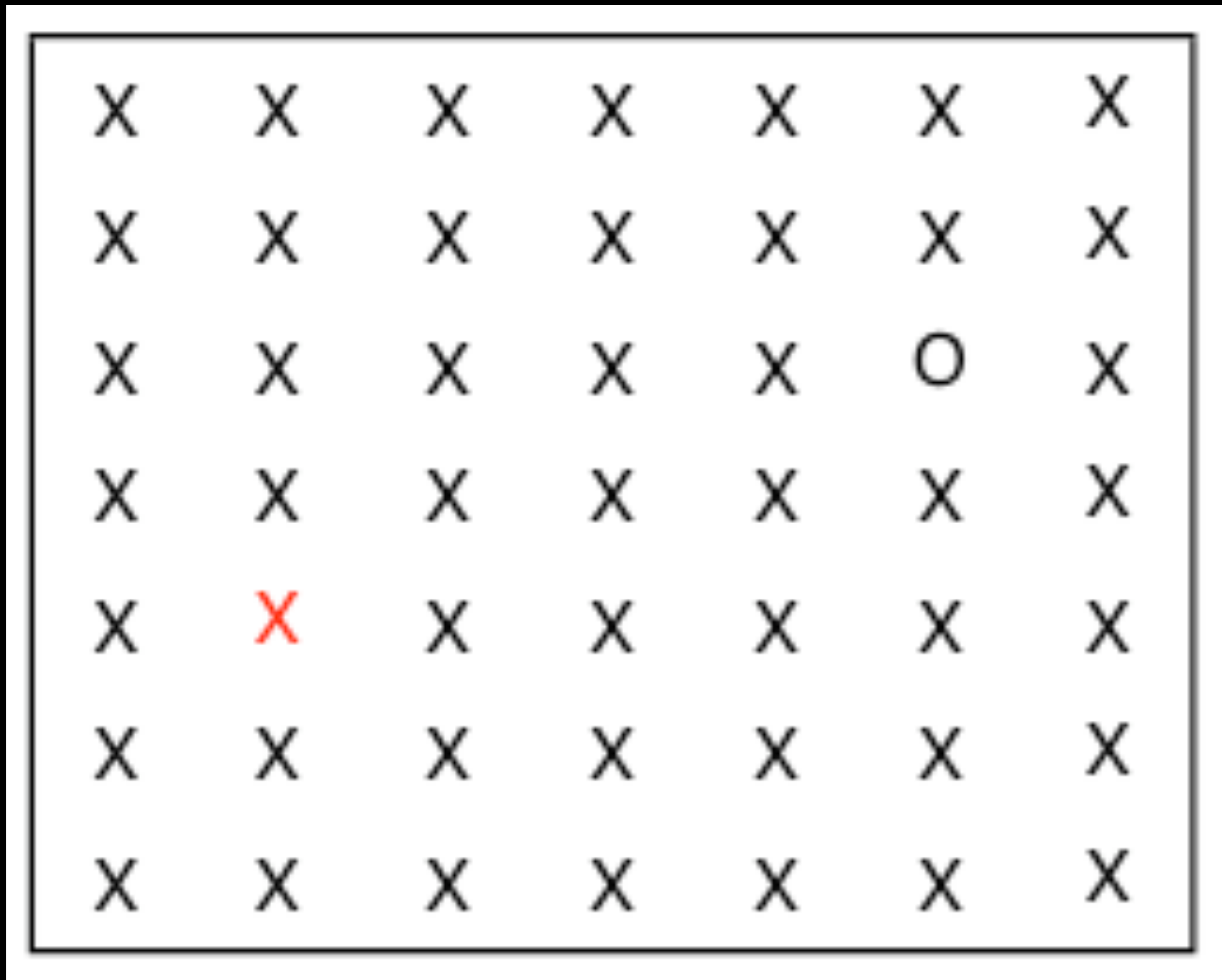
- The stimulus “energy” is insufficient to evoke activity in the visual system and generate conscious perception even when we are paying attention to it
- (e.g., stimulus strength is too week – subthreshold – or presentation time is too brief - subliminal

Attentional factors in emotion recognition

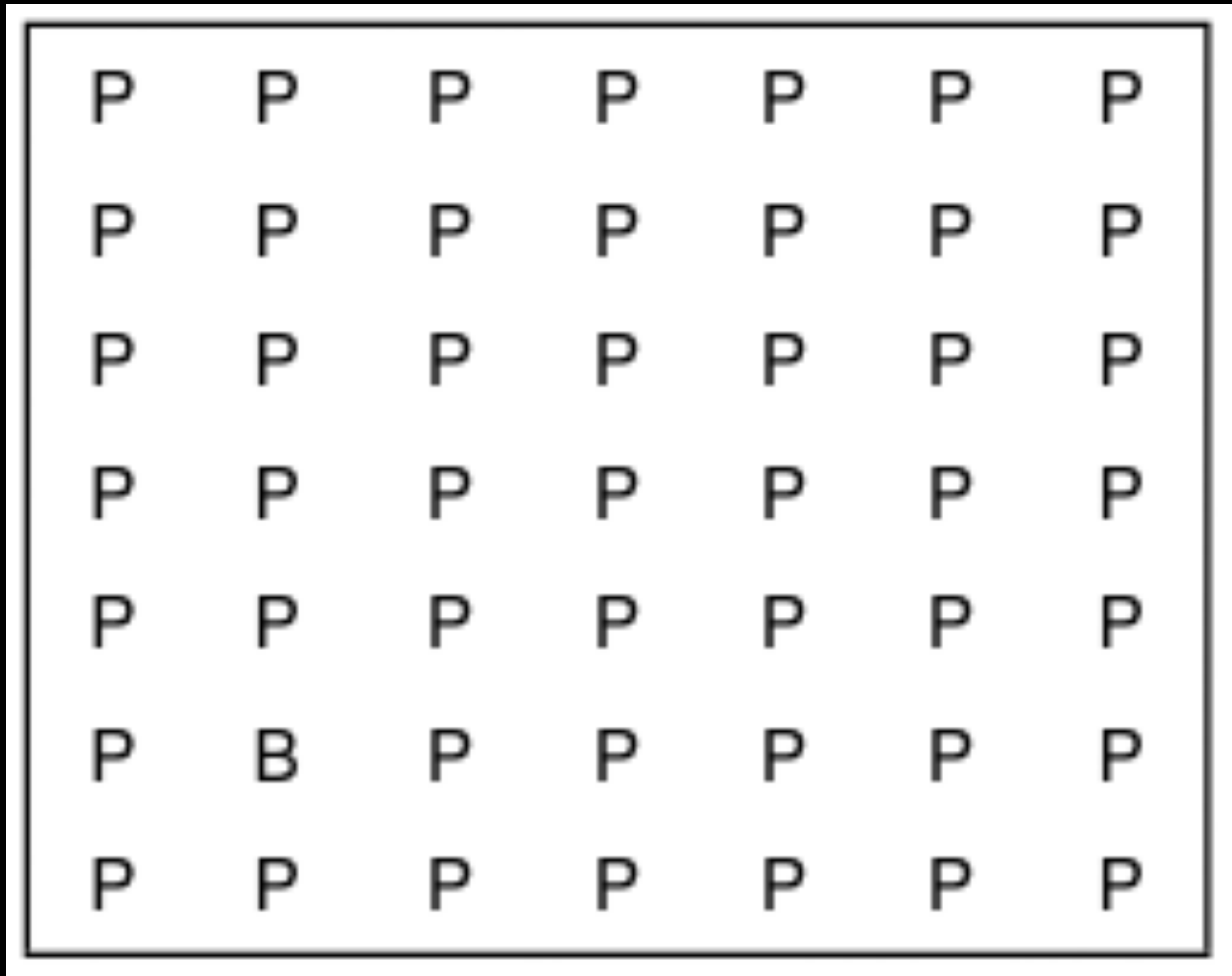


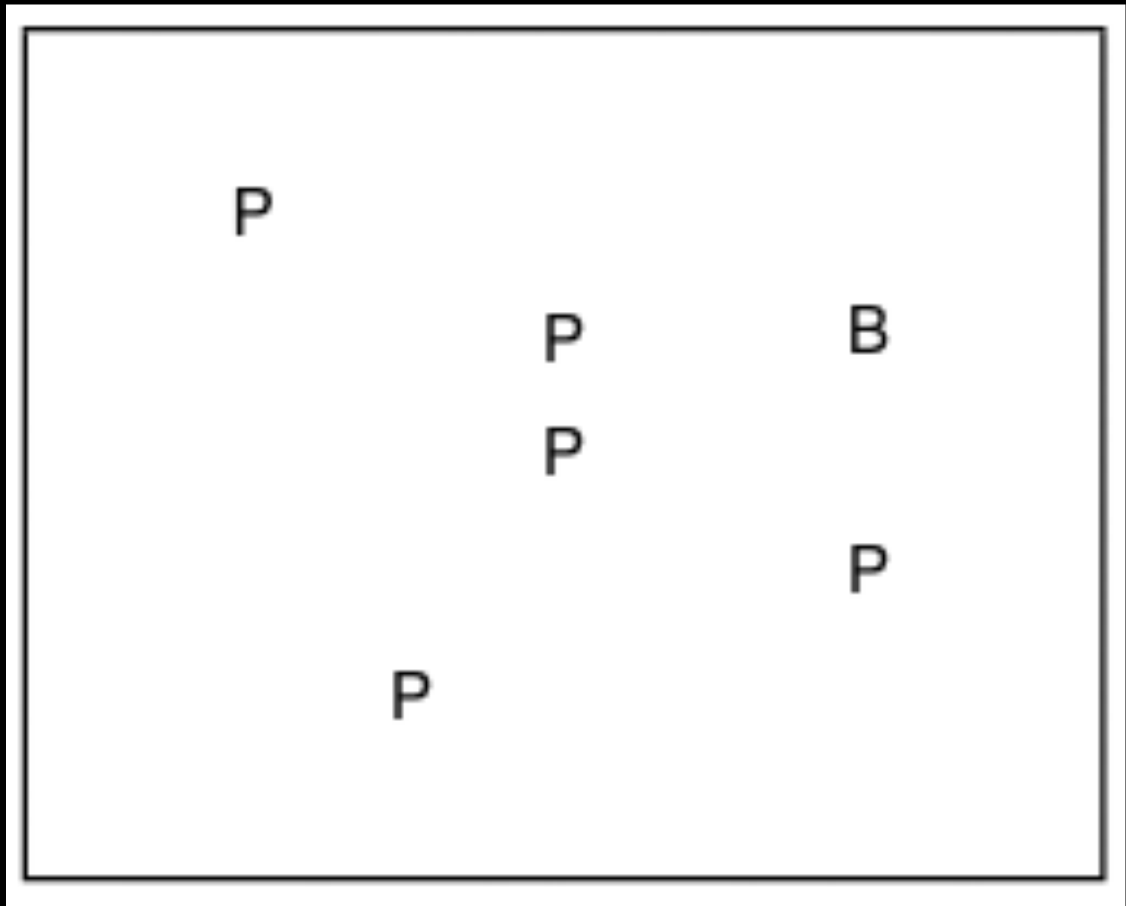


Visual search



Visual search




P
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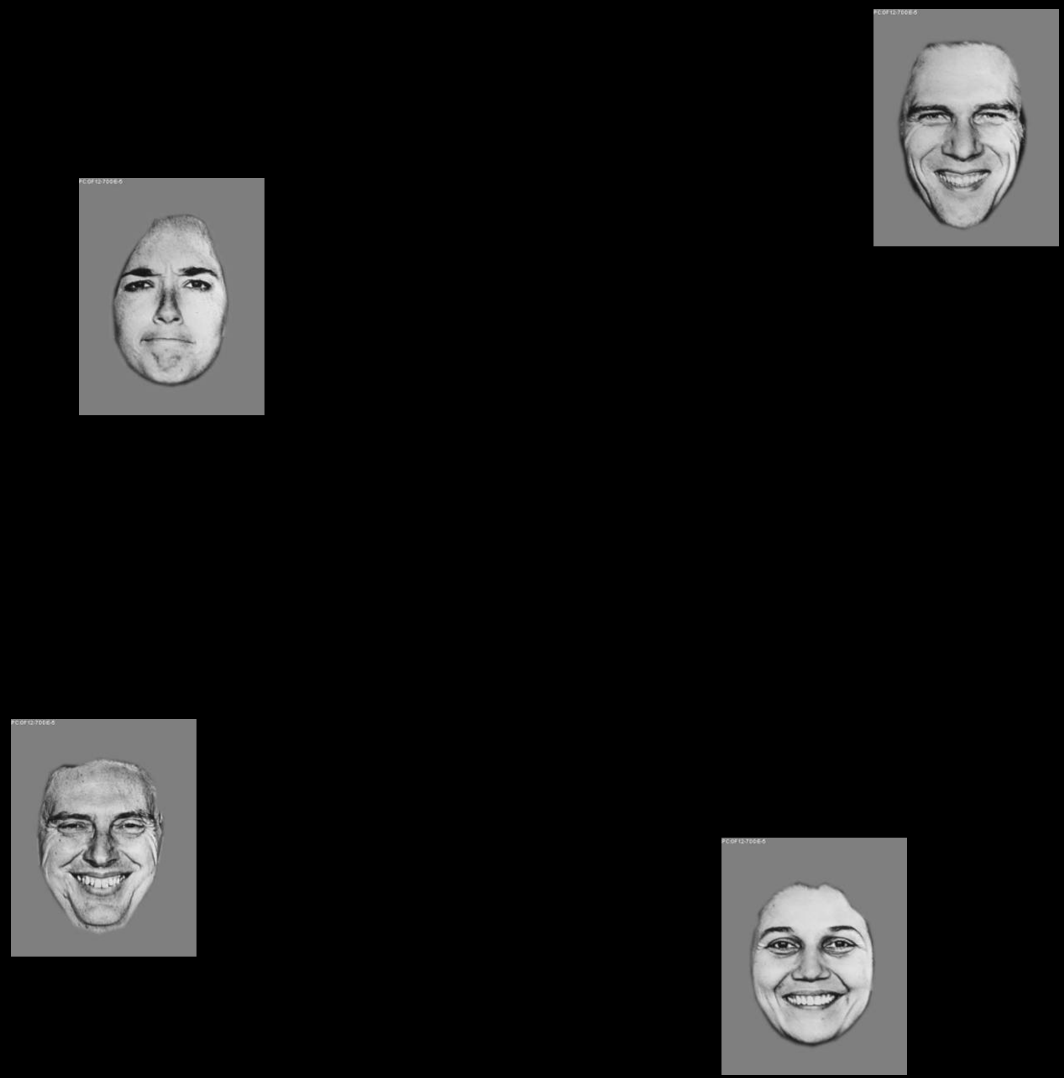
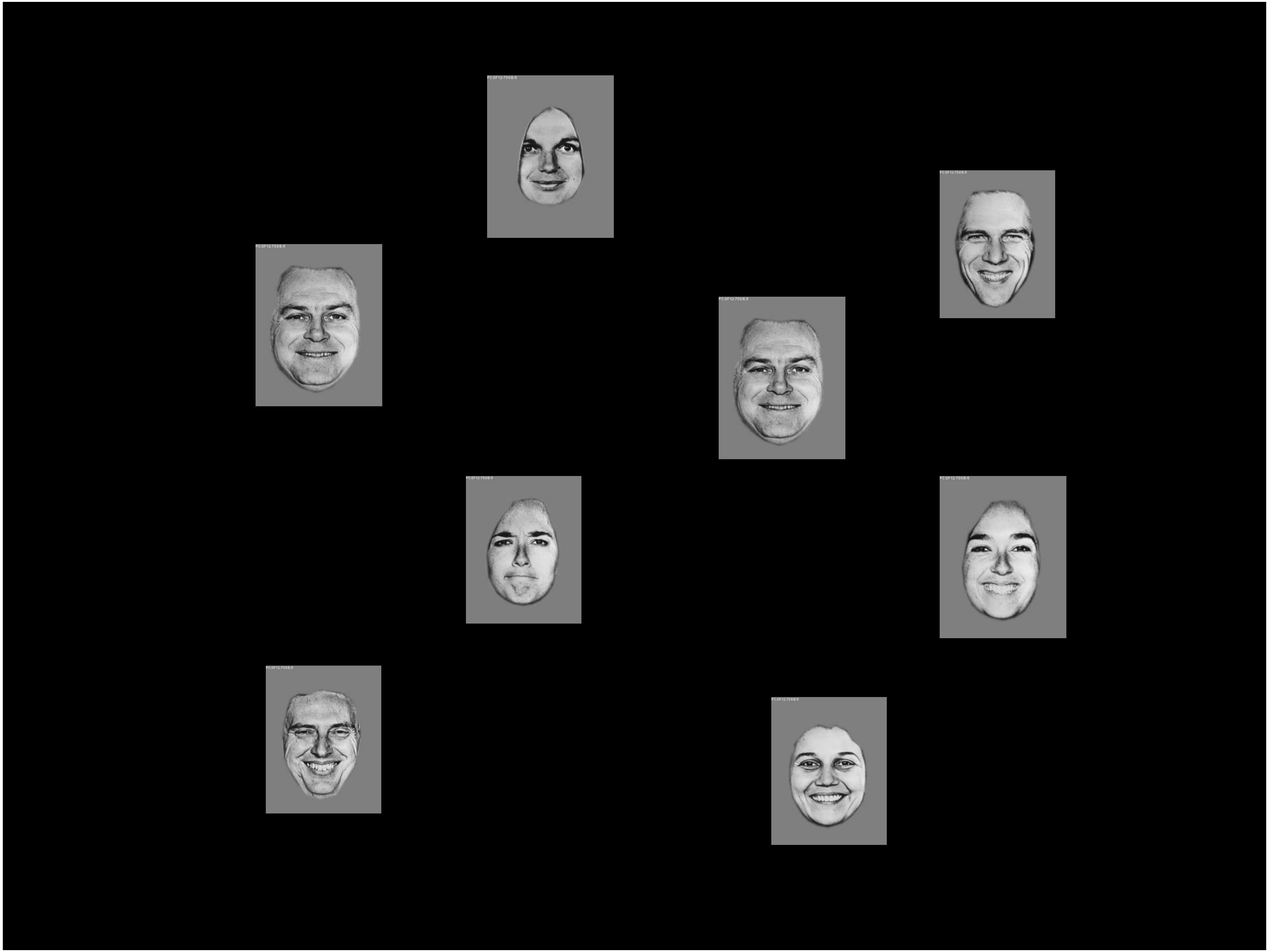
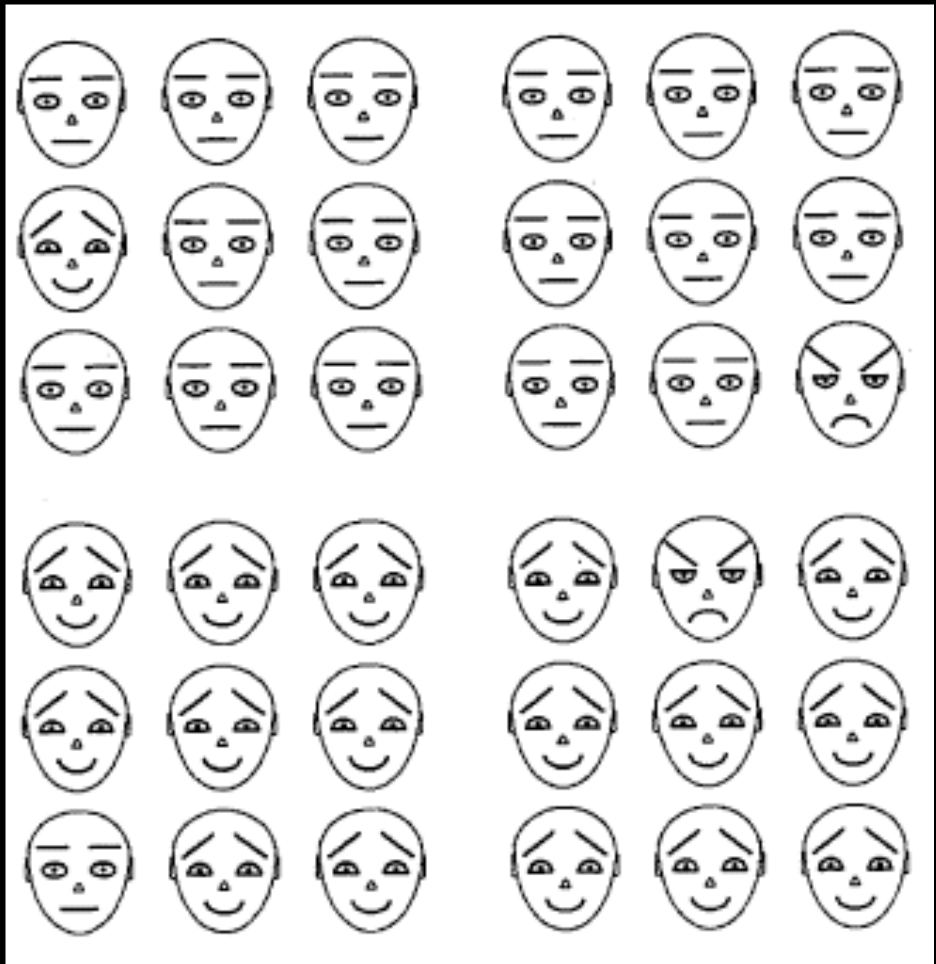
Visual search for emotional stimuli

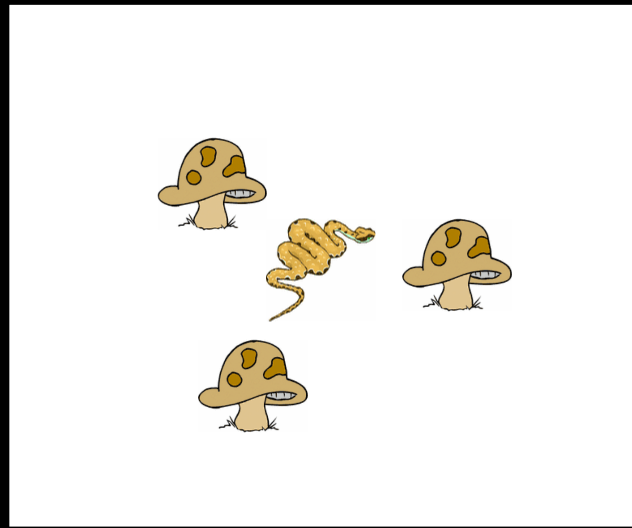


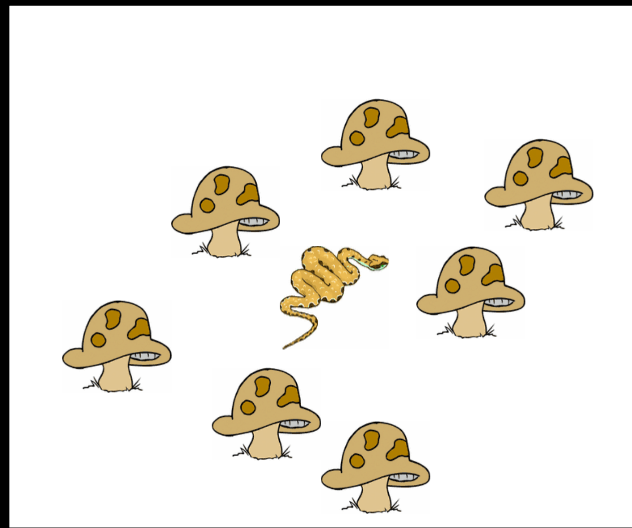


Emotion and attention

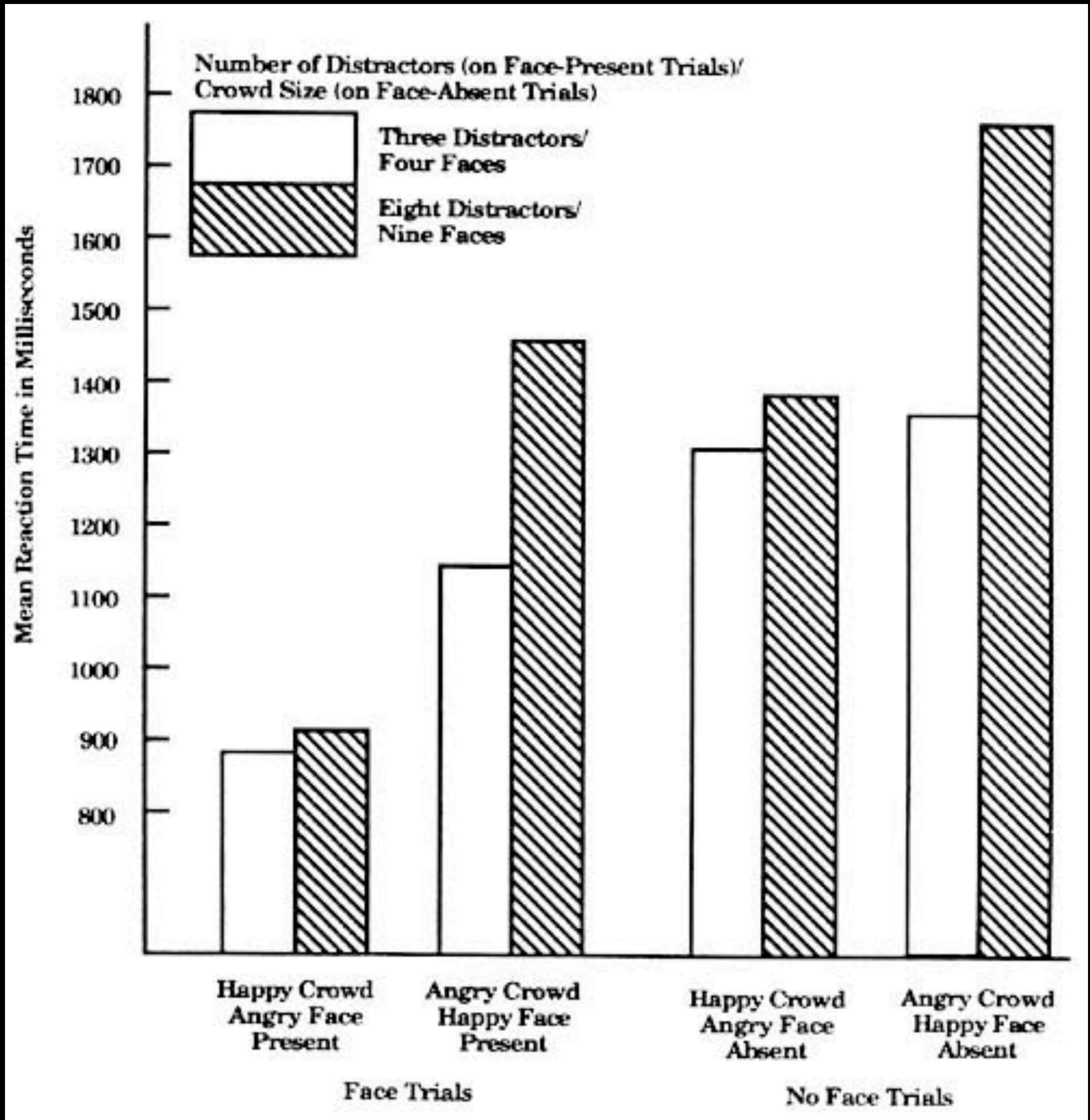
Healthy subjects







Pop-out effect of anger expressions



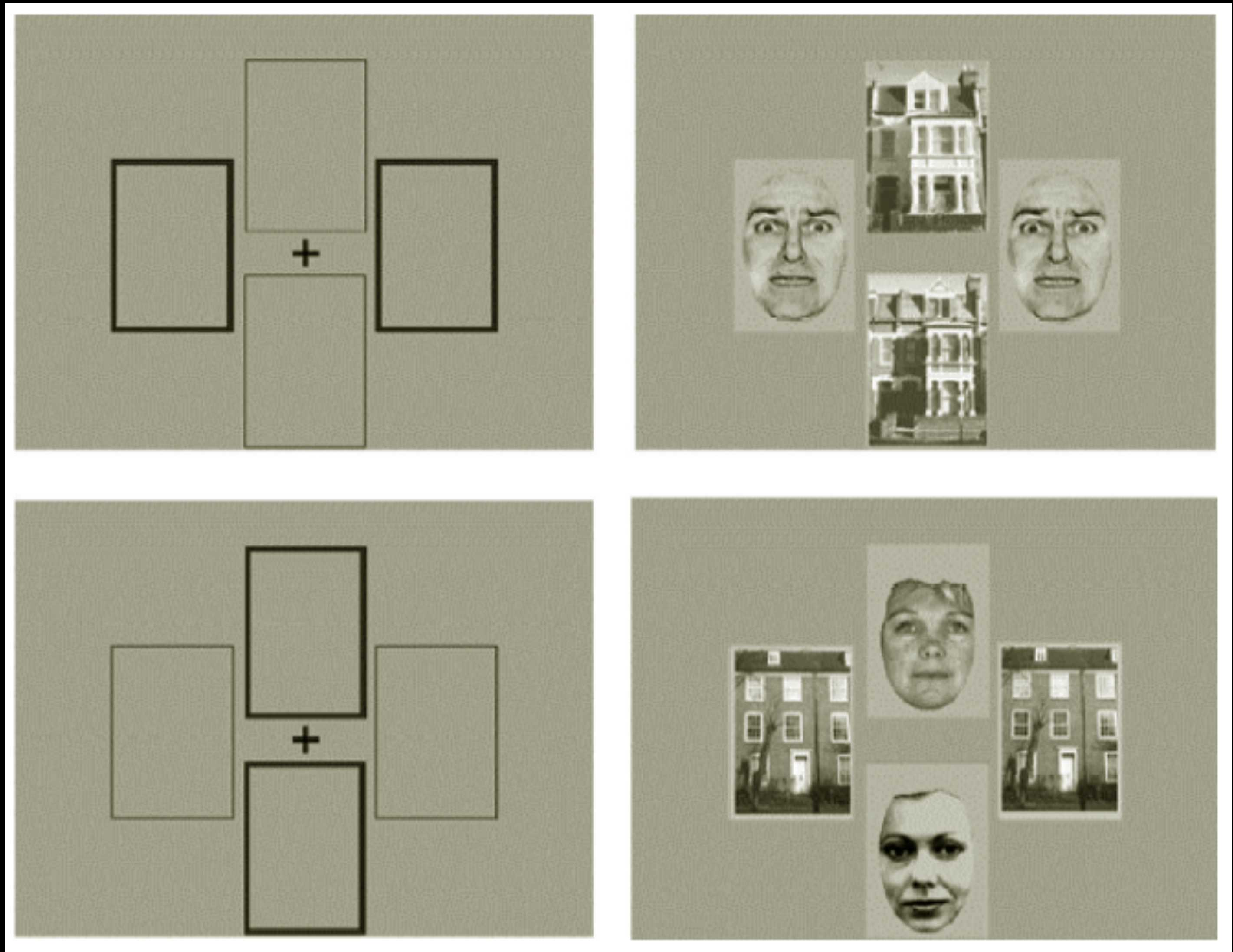
Anger (target) is detected faster than joy

Detection time is not affected by the number of distractors

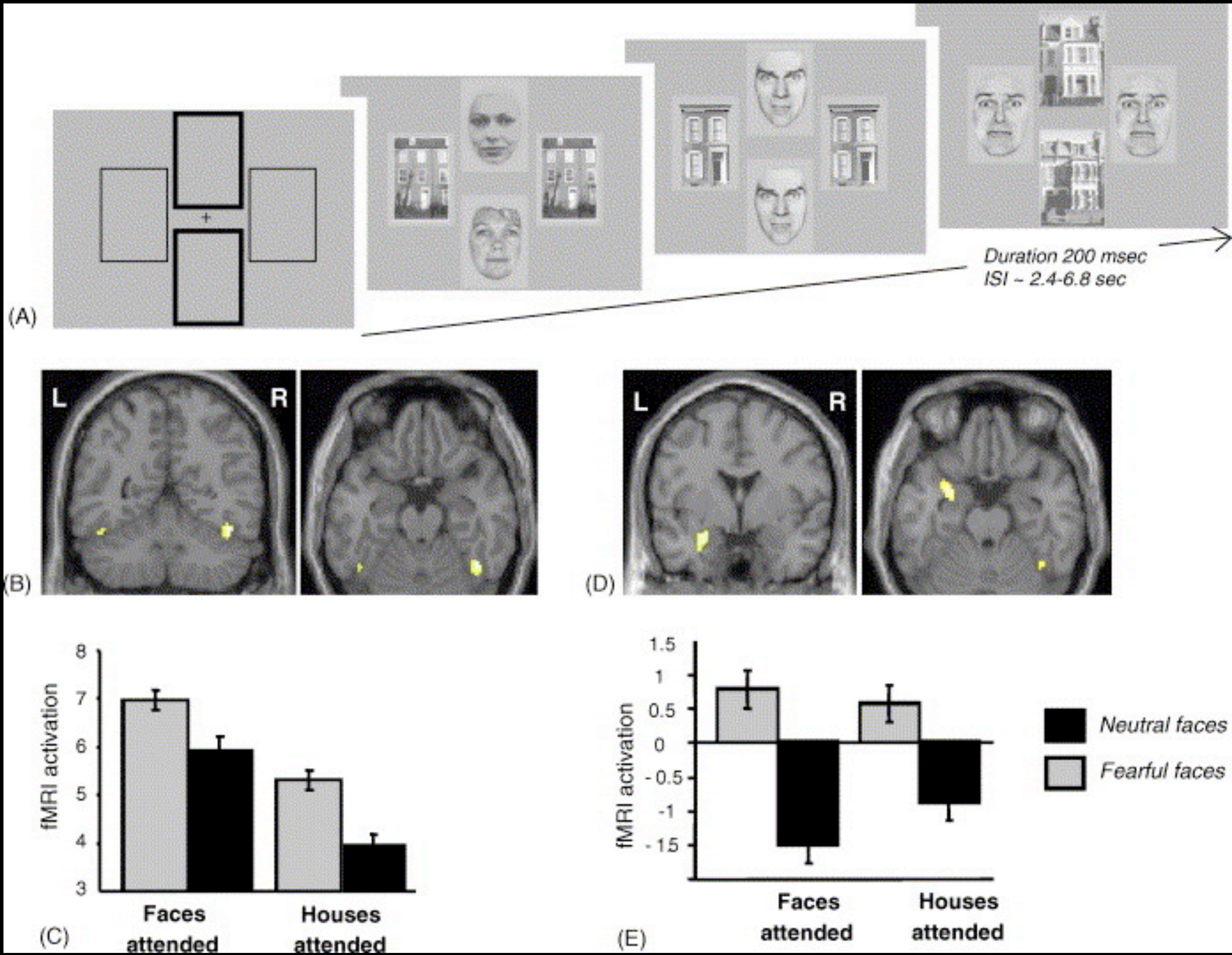
Hansen & Hansen, 1988 J Pers and Soc Psych

Brain structures that process emotions even in the absence of focal attention

fMRI data on healthy subjects



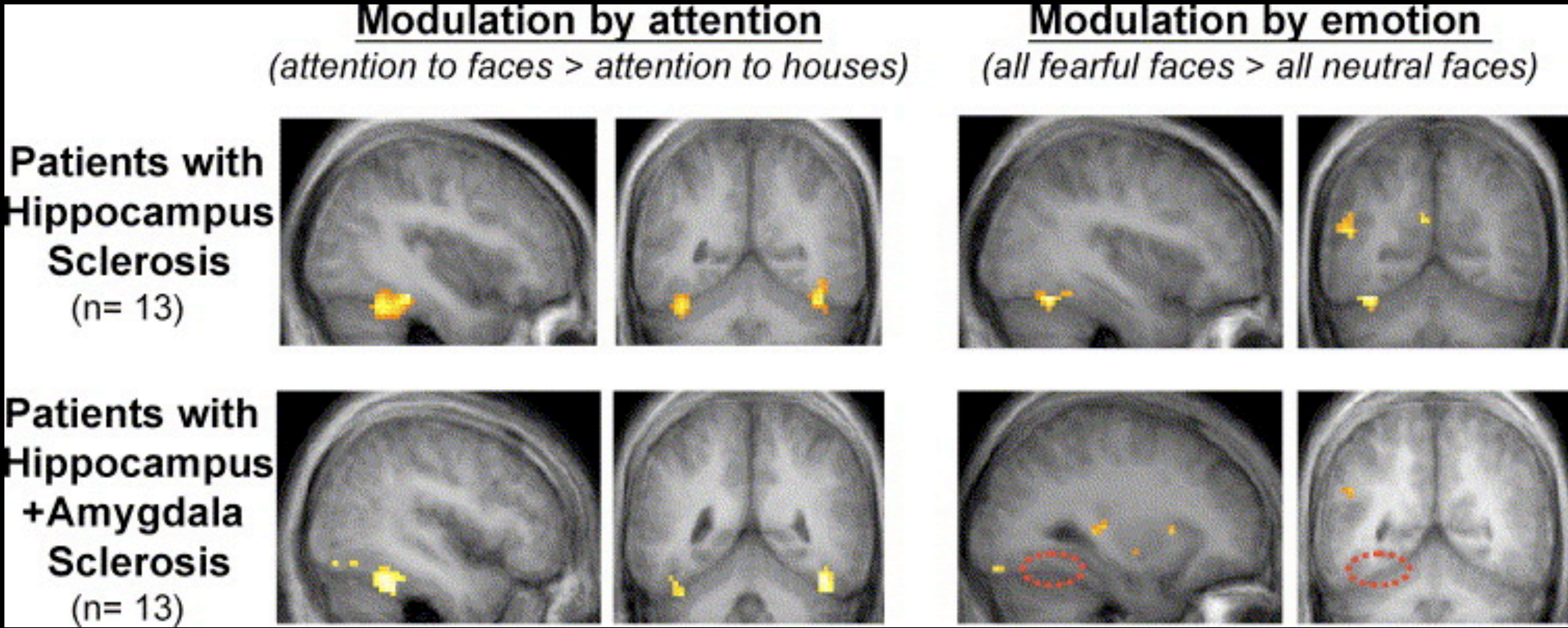
Vuilleumier et al., 2001 Neuron



(A)
Duration 200 msec
ISI ~ 2.4-6.8 sec
(B)
L
R
(D)
L
R
(C)
fMRI activation
8
7
6
5
4
3
Faces attended
Houses attended
(E)
fMRI activation
1.5
1
0.5
0
-0.5
-1
-1.5
Faces attended
Houses attended
Neutral faces
Fearful faces

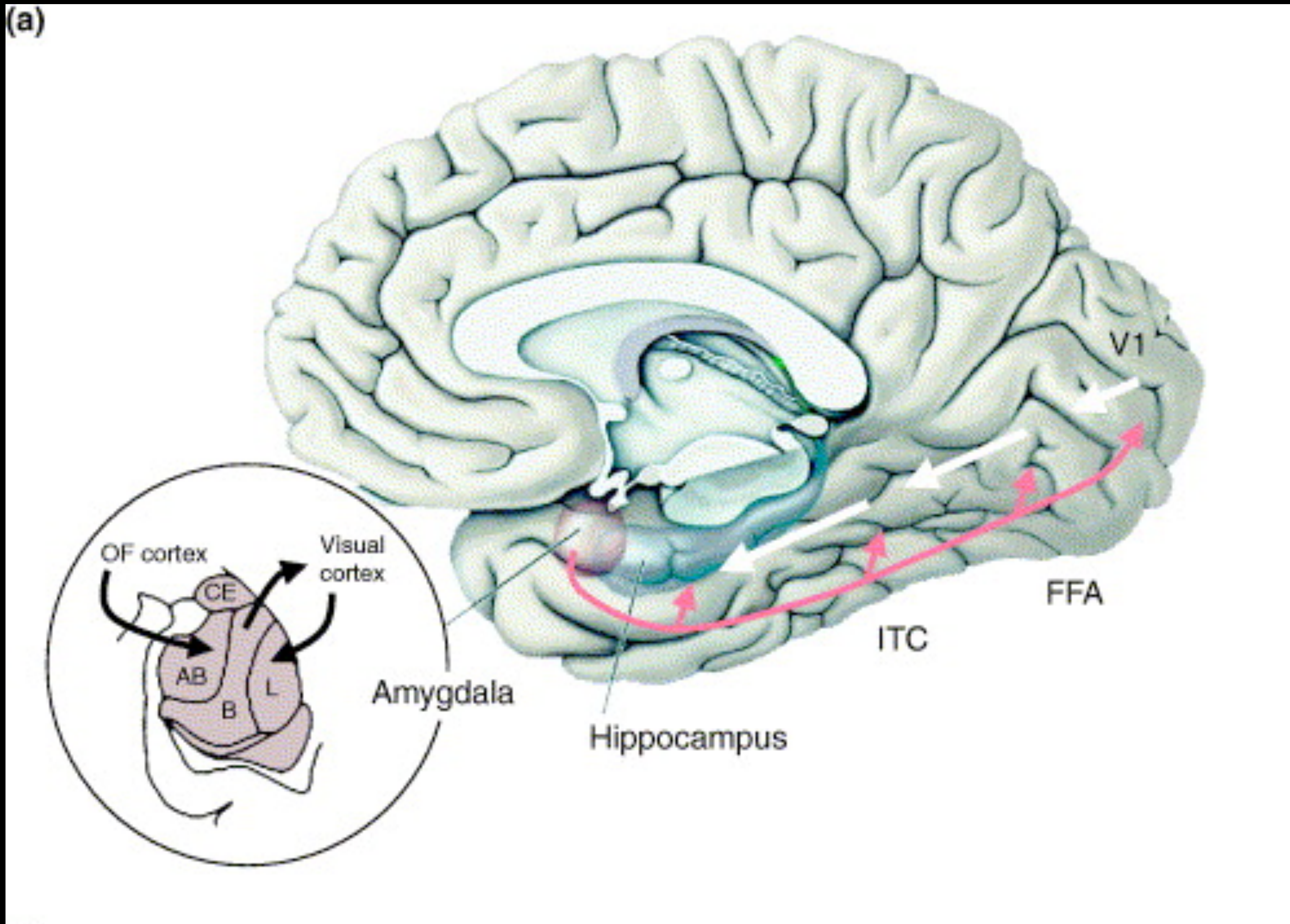
Amygdala interaction visual cortical areas (fusiform gyrus)

fMRI data on neurological patients with focal lesions



Vuilleumier et al.,
2004 Nat Neurosci

Amygdala interaction visual cortical areas (fusiform gyrus)



-Amgdala and attention: a matter of temporal resolution?

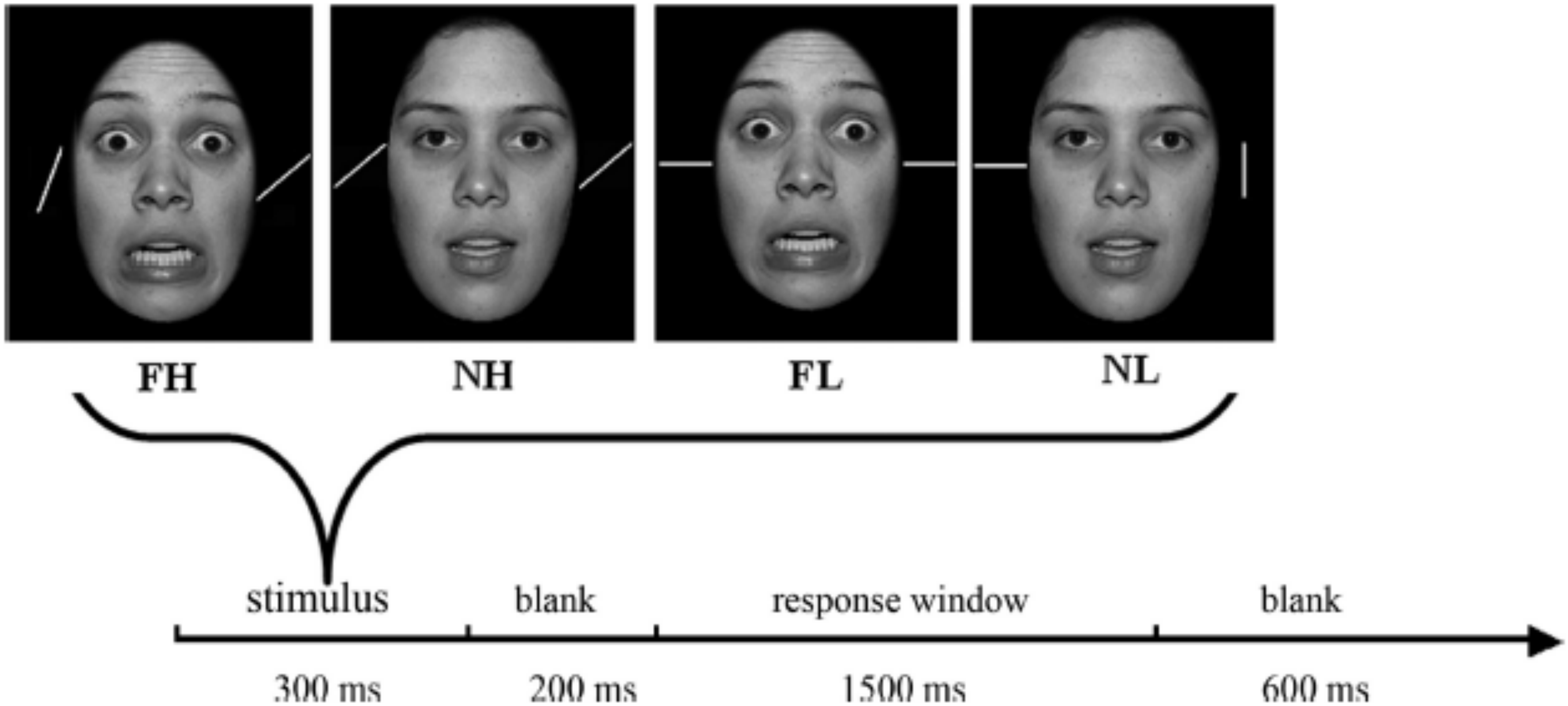


Figure 1. Each stimulus array was presented for 300 ms followed by a 200 ms blank screen. The participant judged whether the bars were parallel during the subsequent 1500 ms response window. The response window was followed by a blank of 600 ms. FH, Fearful expression, high task load; NH. neutral expression, high task load; FL, fearful expression, low task load; NL, neutral expression, low task load.

Luo et al., 2010 J Neurosci

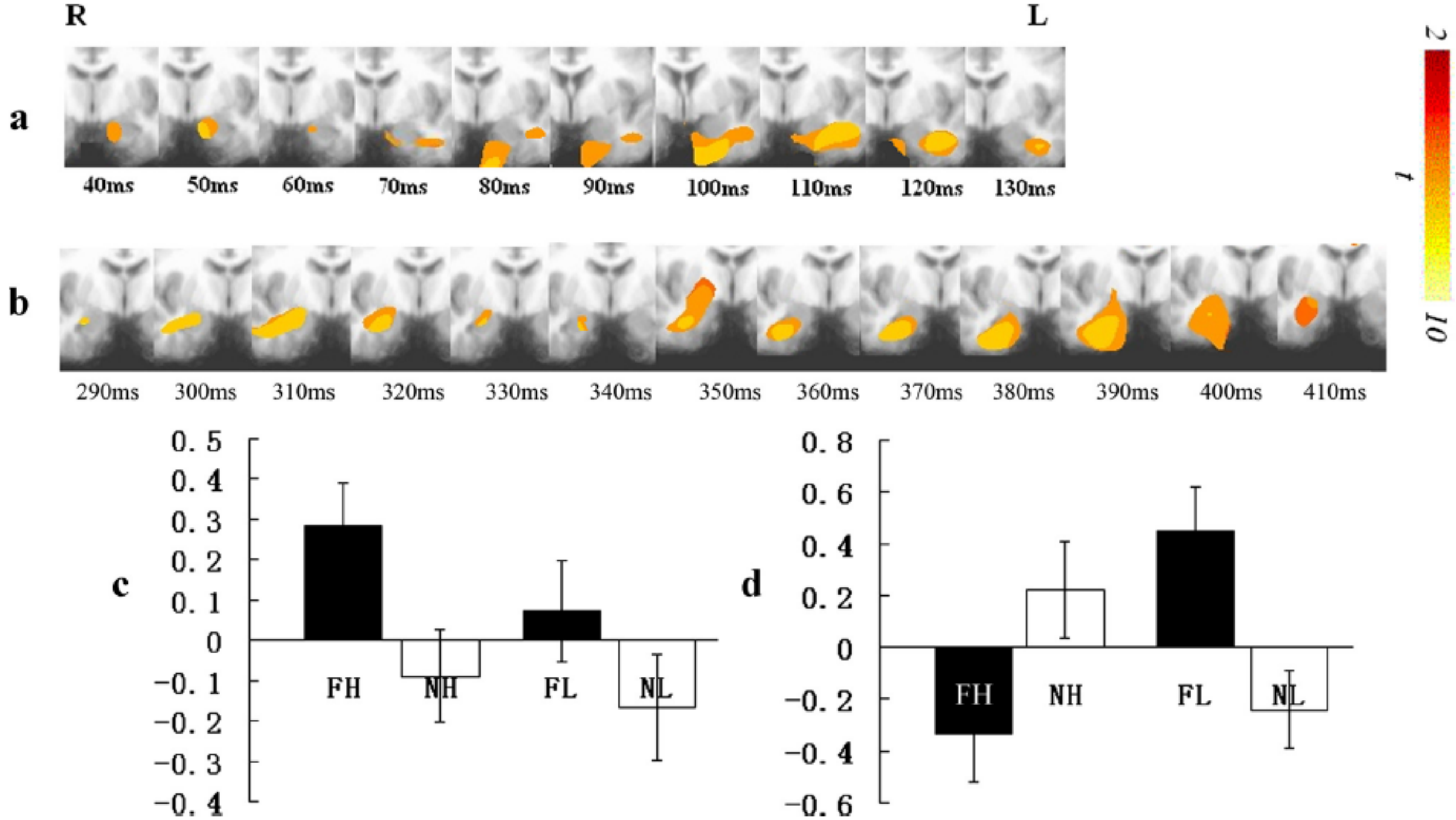


Figure 3. Spatiotemporal profiles of amygdala responses in the gamma band. ***a***, Main effect of distracter emotionality within the left amygdala. R, Right; L, left. ***b***, The task load-by-distracter interaction within the right amygdala. ***c***, Power changes relative to the baseline by condition for the amygdala regions showing a main effect of distracter emotionality. ***d***, Power changes relative to the baseline by condition for the amygdala region showing the task load-by-distracter emotionality interaction. FH, Fearful expression, high task load; NH, neutral expression, high task load; FL, fearful expression, low task load; NL, neutral expression, low task load.

Luo et al., 2010 J Neurosci

Anderson & Phelps 2001 Nature

Attentional blink in patients with amygdala injury

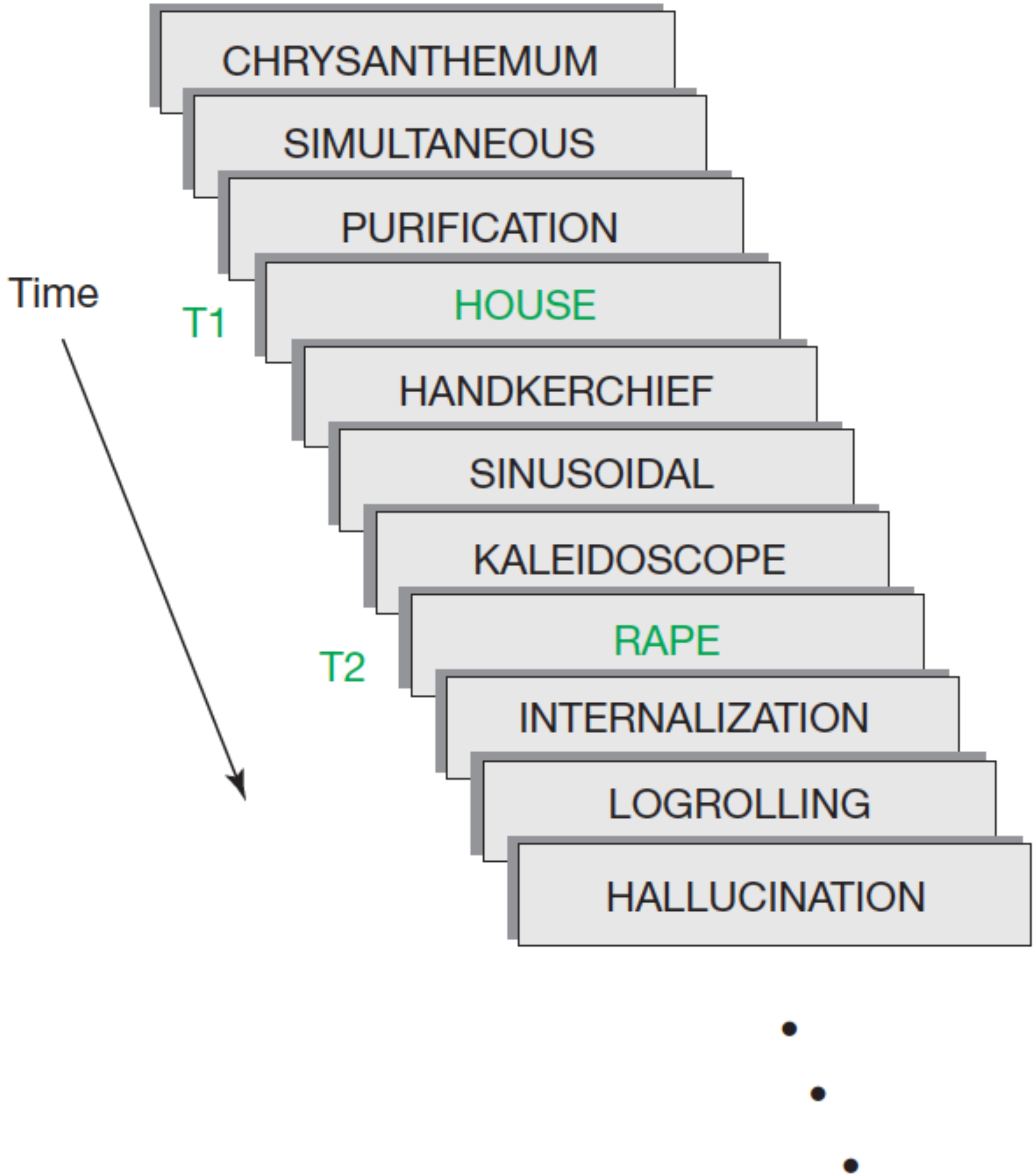


Figure 1 Diagram of the dual-target rapid serial visual presentation (RSVP) task. Fifteen words were briefly presented sequentially in an identical central location, and observers were instructed to ignore words appearing in black (distractors) and indicate the identity of two target words occurring in green. Distractor words were longer in length to ensure sufficient masking of the target events. Responses were made by typing on a computer keyboard immediately after the stimulus sequence. The temporal lag between the first (T1) and second target (T2) was varied. T2 items occurring during early T1–T2 temporal lags (< 4 intervening distractor items, < 600 ms) are susceptible to the attentional blink (attentional blink), an attention-related impairment in perceptual encoding. As attentional resources become more available during late T1–T2 temporal lags (> 4 intervening distractor items, > 600 ms), T2 items are less susceptible to the attentional blink. In the described experiments, the affective value of T2 was manipulated and T2 identification accuracy was the critical measure.

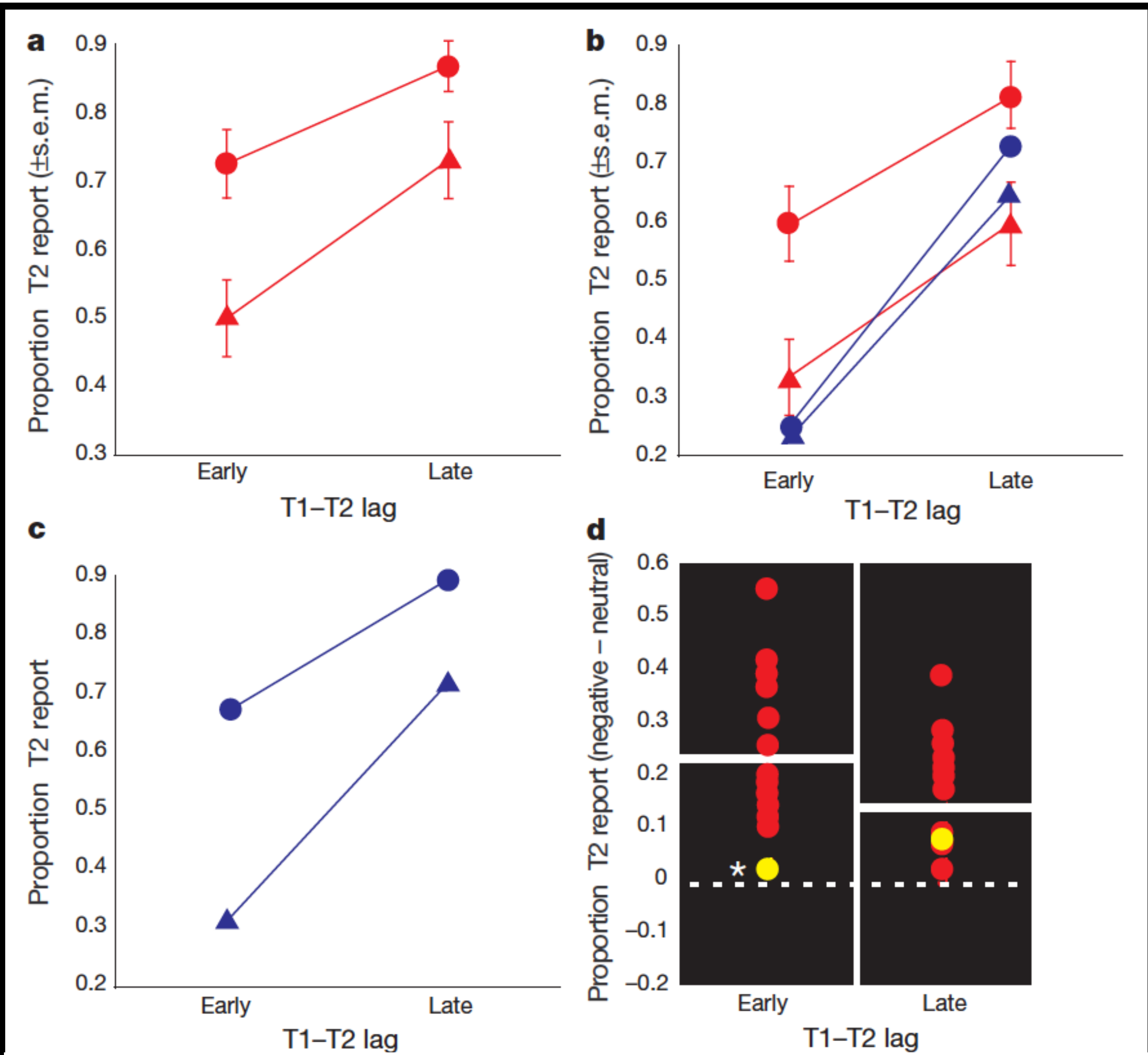


Figure 2 Proportion of T2 items correctly identified at early and late T1–T2 temporal lags. **a**, Controls ($n = 20$; red) showed enhanced identification of negative (circles) compared with neutral (triangles) items during the attentional blink. **b**, S.P. (blue) showed no evidence of enhanced identification of negative (circles) compared with neutral (triangles) items during the attentional blink, and remained significantly impaired when compared with the half of the controls ($n = 10$, in red) that showed similar baseline performance on neutral items. Bars in **a** and **b** show s.e.m. **c**, Impaired affective modulation of perception in S.P. did not generalize to the influence of non-affective manipulations of stimulus salience on perceptual awareness. S.P. exhibits a large benefit for high (circles) compared with low (triangles) salience T2 events. **d**, Distribution of T2 difference scores (negative minus neutral) for controls ($n = 20$) and S.P. at early and late T1–T2 temporal lags. Yellow, S.P.; red, controls; white bar, control mean. In contrast to controls, S.P. did not show a reliable advantage for identifying negative compared with neutral items at early T1–T2 temporal lags, falling significantly outside the control range (controls 22.6%, S.P. 2.1%). Asterisk denotes significant impairment.